



Laplace transforms

Usually In mathematics, a transform refers to a device which changes one kind of function or equation into another kind. In other words, a transform reduces certain problems which we do not know how to solve or which are difficult to solve, into problems which are relatively easier to solve.

The method of Laplace transforms is one of the efficient methods for solving linear differential equations and corresponding initial and boundary value problems. Basically, this method reduces the problem of solving a linear differential equation into an algebraic equation problem whose solution (obtained by usual algebraic methods) is transformed back to obtain the solution of the given problem. The following flow chart illustrates the method of Laplace transforms for the solution of IVPs.

Figure:- A schematic that illustrates the method of Laplace transforms.



Laplace transforms

Laplace transform of a function

If a real-valued function $f(t)$ defined for all $t \geq 0$ is multiplied by e^{-st} and integrated w.r.t. t from 0 to ∞ , a new function of the variable s , say, $\bar{f}(s)$ is obtained where s may be real or complex. The function $\bar{f}(s)$ is called the Laplace transform of the original function $f(t)$ and is denoted by $\mathcal{L}\{f(t)\}$. Thus, we have

$$\bar{f}(s) = \mathcal{L}\{f(t)\} = \int_0^{\infty} e^{-st} f(t) dt.$$

Note it that, the variable of integration t is a dummy variable and can be replaced by any other symbol. The process of finding the Laplace transform of a function is called **Laplace transformation**.



Laplace transforms

Theorem:-

Prove that the Laplacian operator $\mathcal{L}$ is a linear operator, i.e., if C_1 and C_2 are any constants and $f_1(t)$ and $f_2(t)$ are any functions whose Laplace transforms exist, then

$$\begin{aligned}\mathcal{L}\{C_1 f_1(t) + C_2 f_2(t)\} &= C_1 \mathcal{L}\{f_1(t)\} + C_2 \mathcal{L}\{f_2(t)\} \\ &= C_1 \overline{f_1}(s) + C_2 \overline{f_2}(s).\end{aligned}$$

Proof:-

By definition, we have

$$\begin{aligned}\mathcal{L}\{C_1 f_1(t) + C_2 f_2(t)\} &= \int_0^{\infty} e^{-st} [C_1 f_1(t) + C_2 f_2(t)] dt \\ &= C_1 \int_0^{\infty} e^{-st} f_1(t) dt + C_2 \int_0^{\infty} e^{-st} f_2(t) dt. \\ &= C_1 \mathcal{L}\{f_1(t)\} + C_2 \mathcal{L}\{f_2(t)\}.\end{aligned}$$



Laplace transforms

Laplace transforms of some elementary functions

1) Laplace transform of a constant C

Theorem:- Show that

$$\mathcal{L}\{C\} = \frac{C}{s}, \quad s > 0.$$

Proof:- By definition, we have

$$\begin{aligned}\mathcal{L}\{C\} &= \int_0^{\infty} e^{-st} (C) dt \\&= C \int_0^{\infty} e^{-st} dt \\&= C \left| \frac{e^{-st}}{-s} \right|_0^{\infty} \\&= \frac{C}{s}, \quad s > 0.\end{aligned}$$



Laplace transforms

2) Laplace transform of an exponential function

Theorem:- Show that

$$\mathcal{L}\{e^{at}\} = \frac{1}{s-a}, \quad s > a, \quad a \text{ is constant.}$$

Proof:- By definition, we have

$$\begin{aligned}\mathcal{L}\{e^{at}\} &= \int_0^{\infty} e^{-st}(e^{at}) dt \\&= \int_0^{\infty} e^{-(s-a)t} dt \\&= \left| \frac{e^{-(s-a)t}}{-(s-a)} \right|_0^{\infty} \\&= \frac{1}{s-a}, \quad s > a.\end{aligned}$$



Laplace transforms

3) Laplace transform of trigonometric functions

Theorem:- Show that

$$i) \mathcal{L}\{\sin at\} = \frac{a}{s^2 + a^2}, \quad s > 0$$

$$ii) \mathcal{L}\{\cos at\} = \frac{s}{s^2 + a^2}, \quad s > 0$$

Proof:- We have

$$\begin{aligned}\mathcal{L}\{\sin at\} &= \mathcal{L}\left\{\frac{e^{\iota at} - e^{-\iota at}}{2\iota}\right\} \\&= \frac{1}{2\iota}\mathcal{L}\{e^{\iota at}\} - \frac{1}{2\iota}\mathcal{L}\{e^{-\iota at}\} \\&= \frac{1}{2\iota}\left[\frac{1}{s - \iota a} - \frac{1}{s + \iota a}\right] \quad \because \mathcal{L}\{e^{at}\} = \frac{1}{s - a} \\&= \frac{a}{s^2 + a^2}, \quad s > 0.\end{aligned}$$



Laplace transforms

4) Laplace transform of hyperbolic functions

Theorem:- Show that

$$i) \mathcal{L}\{\sinh at\} = \frac{a}{s^2 - a^2}, \quad s > |a|$$

$$ii) \mathcal{L}\{\cosh at\} = \frac{s}{s^2 - a^2}, \quad s > |a|$$

5) Laplace transform of polynomial functions

Theorem:- Show that

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}, \quad n = 0, 1, 2, \dots, \quad \text{and } s > 0.$$



Laplace transforms

	$f(t)$	$\mathcal{L}\{f(t)\} = \bar{f}(s)$
1	1	$\frac{1}{s}, s > 0$
2	t	$\frac{1}{s^2}, s > 0$
3	t^2	$\frac{2}{s^3}, s > 0$
4	$t^n, n = 0, 1, 2, 3, \dots$	$\frac{n!}{s^{n+1}}, s > 0$
5	$t^n, n > -1$	$\frac{\Gamma(n+1)}{s^{n+1}}, s > 0$
6	e^{at}	$\frac{1}{(s-a)}, s > a$
7	$\sin at$	$\frac{a}{s^2+a^2}, s > 0$
8	$\cos at$	$\frac{s}{s^2+a^2}, s > 0$
9	$\sinh at$	$\frac{a}{s^2-a^2}, s > a $
10	$\cosh at$	$\frac{s}{s^2-a^2}, s > a $



Laplace transforms

Example:- Find the following Laplace transforms:

i) $\mathcal{L}\{ae^{bt}\}$

ii) $\mathcal{L}\{a + bt + ct^2\}$

iii) $\mathcal{L}\{6 \sin 2t - 3 \cos 2t\}$

iv) $\mathcal{L}\{3 \cosh 5t - 4 \sinh 5t\}.$

Exercise (Ref. Book No. 1, Problem set 6.1, Question No. 1-8).



Laplace transforms

First shifting theorem for Laplace transforms (s -shifting)

Theorem:- If $\mathcal{L}\{f(t)\} = \bar{f}(s)$, then $\mathcal{L}\{e^{at}f(t)\} = \bar{f}(s - a)$, $s > a$.

Proof:-

By definition

$$\mathcal{L}\{f(t)\} = \bar{f}(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

Then, we have

$$\begin{aligned}\mathcal{L}\{e^{at}f(t)\} &= \int_0^{\infty} e^{-st} [e^{at}f(t)] dt \\ &= \int_0^{\infty} e^{-(s-a)t} f(t) dt \\ &= \bar{f}(s - a).\end{aligned}$$

Hence, the theorem is proved.



Laplace transforms

Example:- Using first shifting theorem, show that

$$i) \quad \mathcal{L}\{e^{at}t^n\} = \frac{n!}{(s-a)^{n+1}}, \quad n = 0, 1, 2, \dots$$

$$ii) \quad \mathcal{L}\{e^{at} \sin \omega t\} = \frac{\omega}{(s-a)^2 + \omega^2}$$

$$iii) \quad \mathcal{L}\{e^{at} \cos \omega t\} = \frac{s-a}{(s-a)^2 + \omega^2}$$

Solution:- 1) Since $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$, $n = 0, 1, 2, \dots$.

Hence, by the first shifting theorem

$$\mathcal{L}\{e^{at}t^n\} = \frac{n!}{(s-a)^{n+1}}, \quad n = 0, 1, 2, \dots$$

Exercise (Ref. Book No. 1, Problem set 6.1, Question No. 33-45).



Laplace transforms

Differentiation of Laplace transforms (multiplication by t^n)

Theorem:- If $\mathcal{L}\{f(t)\} = \bar{f}(s)$, then

$$\mathcal{L}\{t^n f(t)\} = (-1)^n \frac{d^n}{ds^n} \bar{f}(s), \quad n = 1, 2, 3, \dots$$

Proof:- By definition, we have

$$\bar{f}(s) = \int_0^{\infty} e^{-st} f(t) dt.$$

Then by Leibnitz's rule for differentiation under the integral sign, we have

$$\frac{d}{ds} \bar{f}(s) = \frac{d}{ds} \int_0^{\infty} e^{-st} f(t) dt$$



Laplace transforms

$$\begin{aligned}\frac{d}{ds}\bar{f}(s) &= \int_0^{\infty} \frac{\partial}{\partial s} [e^{-st}f(t)] dt \\ &= - \int_0^{\infty} e^{-st} [tf(t)] dt \\ &= -\mathcal{L}\{tf(t)\}\end{aligned}$$

$$\text{or } \mathcal{L}\{tf(t)\} = -\frac{d}{ds}\bar{f}(s)$$

which proves the theorem for $n = 1$. To establish the theorem, in general, we use mathematical induction. Assume that the theorem is true for $n = k$, i.e.,

$$\begin{aligned}\mathcal{L}\{t^k f(t)\} &= (-1)^k \frac{d^k}{ds^k} \bar{f}(s) \\ \text{or } \int_0^{\infty} e^{-st} [t^k f(t)] dt &= (-1)^k \frac{d^k}{ds^k} \bar{f}(s).\end{aligned}$$



Laplace transforms

Then

$$\frac{d}{ds} \int_0^{\infty} e^{-st} [t^k f(t)] dt = (-1)^k \frac{d^{k+1}}{ds^{k+1}} \bar{f}(s).$$

Using Leibnitz's rule, we have

$$\begin{aligned} - \int_0^{\infty} e^{-st} [t^{k+1} f(t)] dt &= (-1)^k \frac{d^{k+1}}{ds^{k+1}} \bar{f}(s) \\ \text{or } \int_0^{\infty} e^{-st} [t^{k+1} f(t)] dt &= (-1)^{k+1} \frac{d^{k+1}}{ds^{k+1}} \bar{f}(s). \end{aligned}$$

Which shows that the theorem holds for $n = k + 1$. Thus the theorem is true for all positive integral values of n .



Laplace transforms

Example:-

Using the above theorem, find

$$i) \mathcal{L}\{t \sin 2t\} \quad ii) \mathcal{L}\{t^2 \sin 2t\}$$

Solution:-

i) Since $\mathcal{L}\{\sin 2t\} = \frac{2}{s^2+4}$, hence, by the above theorem, we have

$$\begin{aligned}\mathcal{L}\{t \sin 2t\} &= -\frac{d}{ds} \left(\frac{2}{s^2+4} \right) \\ &= \frac{4s}{(s^2+4)^2}.\end{aligned}$$



Laplace transforms

Integral of Laplace transforms (division by t)

Theorem:- If $\mathcal{L}\{f(t)\} = \bar{f}(s)$, then

$$\mathcal{L}\left\{\frac{f(t)}{t}\right\} = \int_s^\infty \bar{f}(u) du, \quad \text{provided } \lim_{t \rightarrow 0^+} \left\{\frac{f(t)}{t}\right\} \text{ exists.}$$

Proof:-

By definition

$$\bar{f}(s) = \int_0^\infty e^{-st} f(t) dt.$$

Therefore

$$\bar{f}(u) = \int_0^\infty e^{-ut} f(t) dt.$$



Laplace transforms

Integrating both sides w.r.t. u from s to ∞ , we get

$$\int_s^\infty \bar{f}(u) du = \int_s^\infty \left[\int_0^\infty e^{-ut} f(t) dt \right] du.$$

Interchanging the order of iteration, we have

$$\begin{aligned} \int_s^\infty \bar{f}(u) du &= \int_0^\infty \left[\int_s^\infty e^{-ut} f(t) du \right] dt \\ &= \int_0^\infty f(t) \left| \frac{e^{-ut}}{-t} \right|_s^\infty dt \\ &= \int_0^\infty \frac{f(t)}{t} e^{-st} dt \\ &= \mathcal{L} \left\{ \frac{f(t)}{t} \right\}. \end{aligned}$$



Laplace transforms

Example:- Using the above theorem, find

$$\mathcal{L}\left\{\frac{\sin kt}{t}\right\}.$$

Solution:- Since $\lim_{t \rightarrow 0^+} \left\{\frac{f(t)}{t}\right\} = k$ exist, and

$$\mathcal{L}\{\sin kt\} = \frac{k}{s^2 + k^2}.$$

Hence, we have

$$\begin{aligned}\mathcal{L}\left\{\frac{\sin kt}{t}\right\} &= \int_s^\infty \frac{k}{u^2 + k^2} du \\ &= \left| \tan^{-1} \frac{u}{k} \right|_s^\infty \\ &= \frac{\pi}{2} - \tan^{-1} \frac{s}{k}.\end{aligned}$$

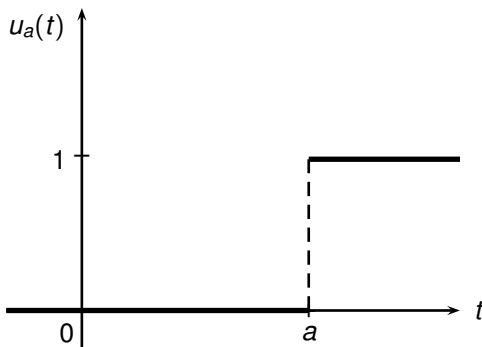


Laplace transforms

Unit step function

The unit step function denoted by $u_a(t)$ is defined as

$$u_a(t) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t \geq a \end{cases} \quad (a \geq 0)$$

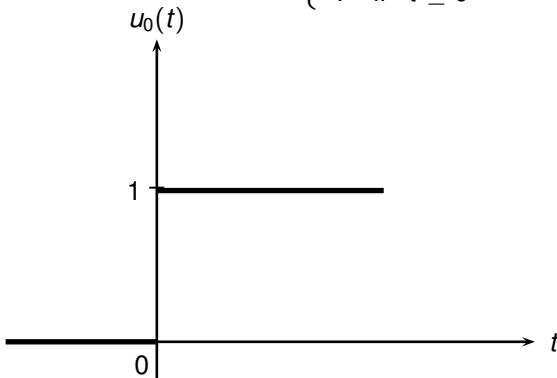




Laplace transforms

Note it that, the unit step function $u_a(t)$ has a jump discontinuity (of size 1) at $t = a$. In-particular, when $a = 0$, i.e.,

$$u_0(t) = \begin{cases} 0 & \text{if } t < 0 \\ 1 & \text{if } t \geq 0 \end{cases}$$





Laplace transforms

The unit step function $u_a(t)$ is a basic building block of various functions and is greatly increases the usefulness of Laplace transform methods.

Note:-

The unit step function is also called Heaviside's unit step function named after the English electrical engineer Oliver Heaviside (1850 - 1925).

Theorem:- Prove that

$$\mathcal{L}\{u_a(t)\} = \frac{e^{-as}}{s}, \quad s > 0.$$

Proof:- Since, we know

$$u_a(t) = \begin{cases} 0 & \text{if } t < a \\ 1 & \text{if } t \geq a \end{cases} \quad (a \geq 0).$$



Laplace transforms

So that, we have

$$\begin{aligned}\mathcal{L}\{u_a(t)\} &= \int_0^{\infty} e^{-st} u_a(t) dt \\&= \int_0^a e^{-st} u_a(t) dt + \int_a^{\infty} e^{-st} u_a(t) dt \\&= \int_0^a e^{-st} (0) dt + \int_a^{\infty} e^{-st} (1) dt \\&= 0 + \int_a^{\infty} e^{-st} dt \\&= \left| \frac{e^{-st}}{-s} \right|_a^{\infty} \\&= \frac{e^{-as}}{s}, \quad s > 0.\end{aligned}$$



Laplace transforms

Second shifting theorem for Laplace transforms (t -shifting)

Theorem:- If $\mathcal{L}\{f(t)\} = \bar{f}(s)$, then

$$\mathcal{L}\{f(t-a)u_a(t)\} = e^{-as}\bar{f}(s), \quad a > 0.$$

Laplace transforms of derivatives

Theorem:- i) If $f(t)$ is continuous on $t \geq 0$ and is of exponential order, i.e.,

$$\lim_{t \rightarrow \infty} e^{-st}f(t) = 0,$$

while $f'(t)$ is piecewise continuous on $t \geq 0$, then

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0) = s\bar{f}(s) - f(0).$$



Laplace transforms

ii) If $f(t)$ and $f'(t)$ are continuous on $t \geq 0$ and are of exponential order, i.e.,

$$\lim_{t \rightarrow \infty} e^{-st} f(t) = 0 = \lim_{t \rightarrow \infty} e^{-st} f'(t)$$

while $f''(t)$ is piecewise continuous on $t \geq 0$, then

$$\mathcal{L}\{f''(t)\} = s^2 \bar{f}(s) - sf(0) - f'(0).$$

iii) If $f(t), f'(t), \dots, f^{(n-1)}(t)$ are continuous on $t \geq 0$ and are of exponential order, i.e.,

$$\lim_{t \rightarrow \infty} e^{-st} f^{(k)}(t) = 0, \text{ for } k = 0, 1, 2, \dots, n-1$$

while $f^{(n)}(t)$ is piecewise continuous on $t \geq 0$, then

$$\mathcal{L}\{f^{(n)}(t)\} = s^n \bar{f}(s) - s^{n-1} f(0) - s^{n-2} f'(0) - \dots - sf^{(n-2)}(0) - f^{(n-1)}(0).$$



Laplace transforms

Proof:- i) By definition, we have

$$\begin{aligned}\mathcal{L}\{f'(t)\} &= \int_0^{\infty} e^{-st} f'(t) dt \\&= \left| e^{-st} f(t) \right|_0^{\infty} + s \int_0^{\infty} e^{-st} f(t) dt \\&= -f(0) + s \mathcal{L}\{f(t)\}, \quad \because \lim_{t \rightarrow \infty} e^{-st} f(t) = 0 \\&= s\bar{f}(s) - f(0)\end{aligned}$$

Example:- Use the above theorem to find the following Laplace transforms:

$$i) \mathcal{L}\{1\} = \frac{1}{s}, \quad ii) \mathcal{L}\{t\} = \frac{1}{s^2}, \quad iii) \mathcal{L}\{e^{at}\} = \frac{1}{s-a}.$$



Laplace transforms

Solution:-

i) Let $f(t) = 1$, therefore $f(0) = 1$ and $f'(t) = 0$. From the formula, we have

$$\mathcal{L}\{f'(t)\} = s\mathcal{L}\{f(t)\} - f(0)$$

$$\mathcal{L}\{0\} = s\mathcal{L}\{1\} - 1$$

$$0 = s\mathcal{L}\{1\} - 1$$

$$\text{or } \mathcal{L}\{1\} = \frac{1}{s}.$$

Laplace transforms of integrals

Theorem:- If $\mathcal{L}\{f(t)\} = \bar{f}(s)$, then

$$\mathcal{L}\left\{\int_0^t f(x) dx\right\} = \frac{\bar{f}(s)}{s}.$$



Laplace transforms

Proof:- Let

$$u(t) = \int_0^t f(x) dx \quad (34)$$

then by the property of definite integrals, i.e., if $F(u) = \int_a^u f(x) dx$ then $\frac{d}{du}F(u) = f(u)$.

From equation (34), we have $u'(t) = f(t)$ and $u(0) = 0$. Then, we have

$$\mathcal{L}\{u'(t)\} = \mathcal{L}\{f(t)\}$$

$$s\mathcal{L}\{u(t)\} - u(0) = \mathcal{L}\{f(t)\}$$

$$\mathcal{L}\{u(t)\} = \frac{1}{s}\mathcal{L}\{f(t)\} \quad \because u(0) = 0$$

$$\text{or } \mathcal{L}\left\{\int_0^t f(x) dx\right\} = \frac{\bar{f}(s)}{s}.$$



Laplace transforms

Example:-

Find the Laplace transforms of the following integrals:

$$i) \int_0^t \cos 3x \, dx, \quad ii) \int_0^t (x^2 - x + e^{-x}) \, dx$$

Solution:-

i) Since $\mathcal{L}\{\cos 3t\} = \frac{s}{s^2+9}$, therefore

$$\begin{aligned} \mathcal{L}\left\{\int_0^t \cos 3x \, dx\right\} &= \frac{1}{s} \left(\frac{s}{s^2+9}\right) \\ &= \frac{1}{s^2+9}. \end{aligned}$$

Exercise (Ref. Book No. 1, Chapter No. 6).



Laplace transforms

Inverse Laplace transforms

If the Laplace transform of a function $f(t)$ is $\bar{f}(s)$, i.e., $\mathcal{L}\{f(t)\} = \bar{f}(s)$. Then $f(t)$ is called the inverse Laplace transform of $\bar{f}(s)$ and we write

$$f(t) = \mathcal{L}^{-1}\{\bar{f}(s)\},$$

where $\mathcal{L}^{-1}$ is the inverse Laplace operator.

Example:- Find the inverse Laplace transforms of the following functions:

$$i) \bar{f}(s) = \frac{1}{s^2} \quad ii) \bar{f}(s) = \frac{1}{s+3} \quad iii) \bar{f}(s) = \frac{s}{s^2 + a^2}$$

Solution:- i) Since $\mathcal{L}\{t\} = \frac{1}{s^2}$, therefore

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$$



Laplace transforms

	$\bar{f}(s)$	$f(t) = \mathcal{L}^{-1}\{\bar{f}(s)\}$
1	$\frac{1}{s}$	1
2	$\frac{1}{s^2}$	t
3	$\frac{1}{s^3}$	$\frac{t^2}{2}$
4	$\frac{1}{s^{n+1}}, n = 0, 1, 2, 3, \dots$	$\frac{t^n}{n!}$
5	$\frac{1}{s^{n+1}}, n > -1$	$\frac{t^n}{\Gamma(n+1)}$
6	$\frac{1}{(s-a)}$	e^{at}
7	$\frac{1}{(s^2+a^2)}$	$\frac{\sin at}{a}$
8	$\frac{s}{(s^2+a^2)}$	$\cos at$
9	$\frac{1}{(s^2-a^2)}$	$\frac{\sinh at}{a}$
10	$\frac{s}{(s^2-a^2)}$	$\cosh at$



Laplace transforms

Note:- In a similar way to that of the above table, we can also look into the inverse version of the previous Laplace transforms theorems.

Convolution

Let $f(t)$ and $g(t)$ are two piecewise continuous functions. Then the convolution of $f(t)$ and $g(t)$ denoted by $f * g$ is defined as

$$f(t) * g(t) = \int_0^t f(u)g(t-u) du.$$

Note it that, since u is a dummy variable of integration. Therefore, $f * g$ is a function of t .

Example:-

Find the following convolution:

$$i) 1 * 1 \quad ii) 1 * \cos t \quad iii) t * e^t$$



Laplace transforms

Solution:- iii) By definition, we have

$$\begin{aligned} t * e^t &= \int_0^t u e^{t-u} du \\ &= e^t \int_0^t u e^{-u} du \\ &= e^t \left| -u e^{-u} - e^{-u} \right|_0^t \\ &= e^t - t - 1. \end{aligned}$$

Properties of convolution

The convolution obeys the following properties:

$$f * g = g * f$$

commutative law of algebra

$$f * (g * h) = (f * g) * h$$

associative law of algebra

$$f * (g_1 + g_2) = f * g_1 + f * g_2$$

distributive law of algebra

$$f * 0 = 0 * f$$



Laplace transforms

However, $1 * g \neq g$, in-general. For example, if $g(t) = t$, then

$$1 * t = \int_0^t 1 \cdot (t - u) du = \frac{t^2}{2}.$$

Convolution theorem

If $\mathcal{L}^{-1}\{\bar{f}(s)\} = f(t)$ and $\mathcal{L}^{-1}\{\bar{g}(s)\} = g(t)$, then

$$\mathcal{L}^{-1}\{\bar{f}(s)\bar{g}(s)\} = f(t) * g(t).$$

Example:-

Using Convolution theorem, find

$$i) \mathcal{L}^{-1}\left\{\frac{1}{s^2(s-a)}\right\} \quad ii) \mathcal{L}^{-1}\left\{\frac{1}{(s+1)(s^2+1)}\right\}$$



Laplace transforms

Solution:-

i) Since $\mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\} = t$ and $\mathcal{L}^{-1}\left\{\frac{1}{s-a}\right\} = e^{at}$. Hence, by the Convolution theorem, we have

$$\begin{aligned}\mathcal{L}^{-1}\left\{\frac{1}{s^2(s-a)}\right\} &= t * e^{at} \\&= \int_0^t u e^{a(t-u)} du \\&= e^{at} \int_0^t u e^{-au} du \\&= e^{at} \left(\frac{1}{a^2} - \frac{te^{-at}}{a} - \frac{e^{-at}}{a^2} \right) \\&= \frac{1}{a^2} (e^{at} - at - 1).\end{aligned}$$



Laplace transforms

Application to initial value problems (IVPs)

An initial value problem is a problem which involves a differential equation (ordinary or partial) along with a condition specified at an initial time $t = 0$. Here, we will restrict ourselves to the discussion for the solution of ordinary differential equations. The following is the procedure to be adopted for the solution of IVPs:

Step (1): Take the Laplace transform of both sides of the given differential equation.

Step (2): Use the initial conditions to obtain an algebraic equation and then solve this equation for the Laplace transform of the required solution.

Step (3): Take the inverse Laplace transform to get the required solution.



Laplace transforms

Example:- Solve the following IVP:

$$y'' + y = 0, \quad y(0) = 1, \quad y'(0) = -1.$$

Solution:- Step (1): Taking the Laplace transform of both sides of the given differential equation, we have

$$\left[s^2 \bar{y}(s) - sy(0) - y'(0) \right] + \bar{y}(s) = 0.$$

Step (2): Using the initial conditions, we get

$$s^2 \bar{y}(s) - s + 1 + \bar{y}(s) = 0$$

$$\text{or } (s^2 + 1) \bar{y}(s) = s - 1$$

$$\text{or } \bar{y}(s) = \frac{s - 1}{s^2 + 1} = \frac{s}{s^2 + 1} - \frac{1}{s^2 + 1}$$



Laplace transforms

Step (3): Taking the inverse Laplace transform, we have

$$\mathcal{L}^{-1}\{\bar{y}(s)\} = \mathcal{L}^{-1}\left\{\frac{s}{s^2 + 1}\right\} - \mathcal{L}^{-1}\left\{\frac{1}{s^2 + 1}\right\}$$
$$y(t) = \cos t - \sin t.$$

Example:- Solve the following IVP:

$$y'' - 9y = -8e^t, \quad y(0) = 0, \quad y'(0) = 10.$$

Solution:- Step (1): Taking the Laplace transform of both sides of the given differential equation, we have

$$\left[s^2\bar{y}(s) - sy(0) - y'(0)\right] - 9\bar{y}(s) = -8\left(\frac{1}{s-1}\right).$$



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Step (2): Using the initial conditions, we get

$$s^2 \bar{y}(s) - s + 1 - 9\bar{y}(s) = \frac{-8}{s-1}$$

$$\text{or } (s^2 - 9)\bar{y}(s) = 10 - \frac{8}{s-1} = \frac{10s - 18}{s-1}$$

$$\text{or } \bar{y}(s) = \frac{10s - 18}{(s-1)(s^2 - 9)} = \frac{10s - 18}{(s-1)(s-3)(s+3)}$$

$$\text{or } \bar{y}(s) = \frac{1}{s-1} + \frac{1}{s-3} - \frac{2}{s+3}$$

Step (3): Taking the inverse Laplace transform, we have

$$\mathcal{L}^{-1}\{\bar{y}(s)\} = \mathcal{L}^{-1}\left\{\frac{1}{s-1}\right\} + \mathcal{L}^{-1}\left\{\frac{1}{s-3}\right\} - \mathcal{L}^{-1}\left\{\frac{2}{s+3}\right\}$$

$$\text{or } y(t) = e^t + e^{3t} - 2e^{-3t}.$$



Laplace transforms

Example:-

Solve the following system of differential equations:

$$y_1' = 2y_1 - 3y_2$$

$$y_2' = y_2 - 2y_1, \quad y_1(0) = 8, \quad y_2(0) = 3.$$

Solution:- Step (1): Taking the Laplace transform of both sides of the given differential equations, we have

$$s\overline{y_1}(s) - y_1(0) = 2\overline{y_1}(s) - 3\overline{y_2}(s)$$

$$s\overline{y_2}(s) - y_2(0) = \overline{y_2}(s) - 2\overline{y_1}(s)$$

Step (2): Using the initial conditions, we get

$$(s - 2)\overline{y_1}(s) + 3\overline{y_2}(s) = 8$$

$$2\overline{y_1}(s) + (s - 1)\overline{y_2}(s) = 3$$



Laplace transforms

Solving the above system using Cramer's rule, we get

$$\overline{y_1}(s) = \frac{5}{s+1} + \frac{3}{s-4}$$
$$\overline{y_2}(s) = \frac{5}{s+1} - \frac{2}{s-4}$$

Step (3): Taking the inverse Laplace transforms, we have

$$y_1(t) = \mathcal{L}^{-1}\{\overline{y_1}(s)\} = 5e^{-t} + 3e^{4t}$$
$$y_2(t) = \mathcal{L}^{-1}\{\overline{y_2}(s)\} = 5e^{-t} - 2e^{4t}$$

Example:- An RLC circuit has a resistance of 10 ohms, a capacitance of 10^{-2} farad, an inductance of 0.5 henry, and an applied voltage $E = 12$ volts. Assume no initial current and no initial charge on the capacitor. Using the method of Laplace transforms, find an expression for the current flowing through the circuit at any time t .



Laplace transforms

Solution:- We know that the differential equation for the charge Q in an RLC circuit is given by

$$\frac{d^2 Q}{dt^2} + \frac{R}{L} \frac{dQ}{dt} + \frac{1}{LC} Q = \frac{1}{L} E(t).$$

Since, we have $R = 10$ ohms, $C = 10^{-2}$ farad, $L = 0.5$ henry, and $E(t) = 12$ volts. Therefore, the above equation becomes

$$\frac{d^2 Q}{dt^2} + 20 \frac{dQ}{dt} + 200Q = 24.$$

Taking the Laplace transform of both sides of the given differential equation, we have

$$\left[s^2 \overline{Q}(s) - sQ(0) - Q'(0) \right] + 20 \left[s \overline{Q}(s) - Q(0) \right] + 200 \overline{Q}(s) = \frac{24}{s}.$$



Laplace transforms

Using the initial conditions: $Q(0) = 0$ and $Q'(0) = 0$, we get

$$(s^2 + 20s + 200)\overline{Q}(s) = \frac{24}{s}$$

$$\begin{aligned}\text{or } \overline{Q}(s) &= \frac{24}{s(s^2 + 20s + 200)} \\ &= \frac{3}{25} \left[\frac{1}{s} - \frac{s + 10}{(s + 10)^2 + 100} \right] - \frac{6}{5} \left[\frac{1}{(s + 10)^2 + 100} \right]\end{aligned}$$

Taking the inverse Laplace transform, we have

$$\begin{aligned}\mathcal{L}^{-1}\{\overline{Q}(s)\} &= \frac{3}{25} \mathcal{L}^{-1}\left\{ \frac{1}{s} - \frac{s + 10}{(s + 10)^2 + 100} \right\} - \frac{6}{5} \mathcal{L}^{-1}\left\{ \frac{1}{(s + 10)^2 + 100} \right\} \\ Q(t) &= \frac{3}{25} (1 - e^{-10t} \cos 10t) - \frac{3}{25} e^{-10t} \sin 10t \\ &= \frac{3}{25} - \frac{3}{25} e^{-10t} (\cos 10t + \sin 10t).\end{aligned}$$



Laplace transforms

The Laplace transform method offers several advantages, particularly in the analysis of linear systems and differential equations:

- 1 Solving a nonhomogeneous ODE does not require first solving the corresponding homogeneous ODE.
- 2 Initial conditions are incorporated directly into the transformed equations, eliminating the need for separate computations.
- 3 Complicated inputs $f(t)$ (right sides of linear ODEs) can be handled very efficiently.
- 4 The method effectively handles functions with discontinuities (e.g., step functions) and impulses (e.g., Dirac delta functions), which are difficult to manage in the time domain.
- 5 It is widely used in control systems, circuit analysis, and signal processing because it provides a systematic framework for analyzing system behavior.



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